

Integrated Prediction of Mechanical Behavior for the Non-Aligned Fiber Composites With Experimental Validation

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This paper aims to present an integrated multi-scale method for predicting the anisotropic and nonlinear elasto-plastic behavior of short glass fiber-reinforced polymer (GFRP) materials typically produced by injection molding. The proposed method combines injection molding and microstructure together, with considering the nonaligned fibers and their corresponding anisotropy, to semi-analytically estimate the local effective mechanical properties at every point of GFRP. Micro-computed tomography measurement and injection molding simulation are used to obtain the fiber orientation tensor. The two-step mean-field homogenization method is applied to calculate the mechanical behaviors of the PA66GF30 GFRP with distributed-orientation fibers based on the fiber orientation tensor. Reverse engineering is used to obtain the optimized parameters of J_2 -plasticity and Tsai-Hill three-dimensional transversely isotropic stain-based failure criterion. Moreover, the integral mapping method can complete the transformation of the fiber orientation tensor from injection simulation to structure simulation model. The proposed integrated approach with the optimized parameters is verified by predicting the ring samples' behavior from injection plates. The results from this investigation are expected to provide some design guidelines for GFRP composites. [DOI: 10.1115/1.4047744]

Keywords: mechanical behavior, fiber-reinforced polymer, short non-aligned fiber, experimental validation, integrated prediction

1 Introduction

The composite component is frequently used in electronic and automotive industries to replace the traditional steel components since it has excellent performance in lightweight and high degree of integration. Glass fiber-reinforced polymers (GFRPs) have attracted a significant amount of attention for the good capacity of manufacturability and low cost when produce automotive and electronic parts using injection molding and free-shape design. From a practical point of view, blending glass fibers (GFs) with engineering plastics is an effective way to obtain desired mechanical and physical properties.

The GFRP components that consist of discontinuous fiber-reinforced composites are produced by injection molding techniques. During this process, the fibers are dispersed due to the resin flow and thus have various fiber orientation distributions (FODs). The material properties of GFRP highly depend on the FOD which are locally heterogeneous in an injected component. In this case, the fiber orientation significantly affects the mechanical strength of the GFRP composites. The design of GFRP composites involves using a complex structure with various design characteristics, such as holes, slots, bosses, and molded-in treads, for assembling with other parts. These design characteristics affect the flow state of molten resin which contributes the formation of complicated fiber distributions during the injection molding process. Also, once external loads are applied to the parts, complicated stress distribution is then will be induced since the stress concentration and interference from the design elements will directly affect the stiffness

and strength of the parts. Therefore, it is necessary to examine both the stress and fiber distributions at the early step of designing plastic parts [1]. However, accurate and reliable prediction of GFRP properties is a challenging subject since that complex fiber orientation from injection molding producing nonlinear and strong anisotropy [2]. Fiber mass fractions of up to 30% can be found in the automotive industry and electronic components, where the fibers themselves have a high stiffness with a linear mechanical response. However, the polymeric matrix dispersed in the composites is predominantly nonlinear.

Polymer materials reinforced with glass fiber have received tremendous attention in both scientific and industrial communities due to their extraordinary enhanced properties, such as lower weight, higher toughness, and higher strength characteristics. Many experimental, numerical, and theoretical studies on these glass fiber-reinforced composites have been carried out to investigate the fiber distributions, material models, and mechanical behaviors [3–6].

Fiber distributions have been extensively studied since it is very important for understanding the changes in the mechanical behaviors of plastics fabricated by injection molding [7]. It has been researched extensively of the relationships between the mechanical behaviors and the fiber orientations [1,8–10], the fiber length [11,12], and the injection molding conditions [7,13,14]. Various related studies, such as the modeling and analysis of fiber distributions [15,16] and the experiments on the measurement of fiber orientations [17,18], have been carried out for injection molded parts. Micro-computed tomography (μ CT) has been popularly used to analyze the complexity of fiber distributions [19,20]. The performance of the fiber-reinforced polymers at macroscopic and microscopic scales has been studied by micro-tomography [17,21]. However, most studies focused on the mechanical behaviors of the polymer are under simple stress state. Limited research has been conducted on the mechanical response of polymers under

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complex stress state. Most studies have not put effort on investigating the relationships between the nonaligned fiber orientations and the nonlinear behavior using the integrated micro and macro approach.

Mean-field homogenization (MFH) methods and representative volume element (RVE) are widely used in the research of material models and mechanical behavior. RVE can be understood as a kind of micro-unit, and the equivalent mechanical behavior on the macro scale can be obtained through its structure simulation [22,23]. MFH methods are a multi-scale analytical algorithm by using the mean-field homogenization method, which equalizes composite materials into homogeneous materials to calculate the effective mechanical properties of multiphase materials. The Halpin-Tsai algorithm is proposed in 1969 to describe the behavior of the fiber composite material [24,25], and it is used to calculate the elastic modulus [26], which considers the fiber volume, fiber orientation, and fiber length. Based on equivalent inclusion of Eshelby's theory, Mori and Tanaka [27] proposed a Mori-Tanaka algorithm, which considering the interaction of dispersed components in multiphase composites, to effectively solve the elastic modulus of heterogeneous materials in 1973. Furthermore, Benveniste [28] simplified and clearly annotated the Mori-Tanaka (M-T) model. The functional expression of aspect ratio (AR), volume factor, and fiber orientation to the equivalent elastic modulus are deduced and compared with the experiments by using the M-T method. It concludes that the Mori-Tanaka algorithm could better predict the elastic modulus of short-fiber composites [29,30]. For the nonlinear behavior, little literature [31,32] has constructed the constitutive model to calculate the nonlinear mechanical properties within a certain strain range. Based on the so-called pseudo-grains presented RVE, Kammoun et al. [33], Doghri et al. [34], and Kammoun et al. [35] proposed a general two-step incremental formulation and the corresponding numerical algorithms, which are able to deal with any rate-independent model for any phase as well as cyclic or otherwise non-proportional loadings. Compared with the micro RVE solution, the macro elastic-plastic properties of composites can be obtained more quickly through the analytical calculation with M-T and two-step algorithm [36–38]. As mentioned above, researchers have done basic studies on the material properties of simple GFRP samples with aligned fibers. Yet, few researchers consider the case of no-aligned fiber caused by the injection molding process, such as the agglomeration phenomenon of fiber layering.

The mechanical properties of injection molded short glass fiber-reinforced polyamide 6,6 with 30% w.t. fibers (PA66GF30) are investigated in this paper. Discontinuous fiber-reinforced

PA66GF30 requires a more comprehensive understanding of its performance for its better application in lightweight and precise product design [39,40] since its wide application in the automobile industry and machinery parts and component manufacture. An integrated method for the nonlinear mechanical behavior assessment of GFRP is developed. It is a fully integrated numerical approach linking the injection simulation to the estimation of the nonlinear behavior. The integrated method is made of three modules. First module (Secs. 2 and 3) is the injection specimens and tensile test to obtain the mechanical behavior and μ CT scanner to provide the orientation tensor of fiber distribution. Second module (Sec. 4) is the micro or macro estimation of the local effective properties of the GFRP material at every point, based on the orientation tensor and anisotropic nonlinear material model. Third module (Secs. 5 and 6) is to compute the mechanical behavior of the part over one cycle for each applied load level and to verify the integrated methods with experiments. With the goal of making the approach usable in an industrial context, the calculation chain was based on developed PYTHON script and different commercial software (MOLD-FLOW®, DIGIMAT®, and ABAQUS®). The identification of the model parameters and the validation of the methodology relied on the tensile tests and μ CT scanner data.

2 Injection Processing and Tensile Test of Specimens

2.1 Injection Processing. Discontinuous fiber-reinforced thermoplastic composites are increasingly used in industry when enhanced mechanical properties and lightweight and high accuracy are required. Moreover, they can be easily formed into complex-shaped structures by using the injection molding process. With the high automation level, high efficiency, and process versatility, the injection molding process is widely used in manufacturing technologies for mass production of plastic parts [41].

The investigated material will be commercial PA66GF30, provided by Kingfa Plastics (China), which contains 30% mass or 18.88% volume of short glass fibers. The main geometrical features of the investigated injected flat plate are rectangular shape with dimensions of 150 mm $\times$ 100 mm and 1.65 mm thickness, as shown in Fig. 1.

The injection molding equipment in use is the HAITIAN MA1600/540, with a clamping force of 122 kN. The material was dried at 100 °C for 4 h, the melt temperature was 280 °C for PA66, the mold temperature was 75 °C, and the average cycle

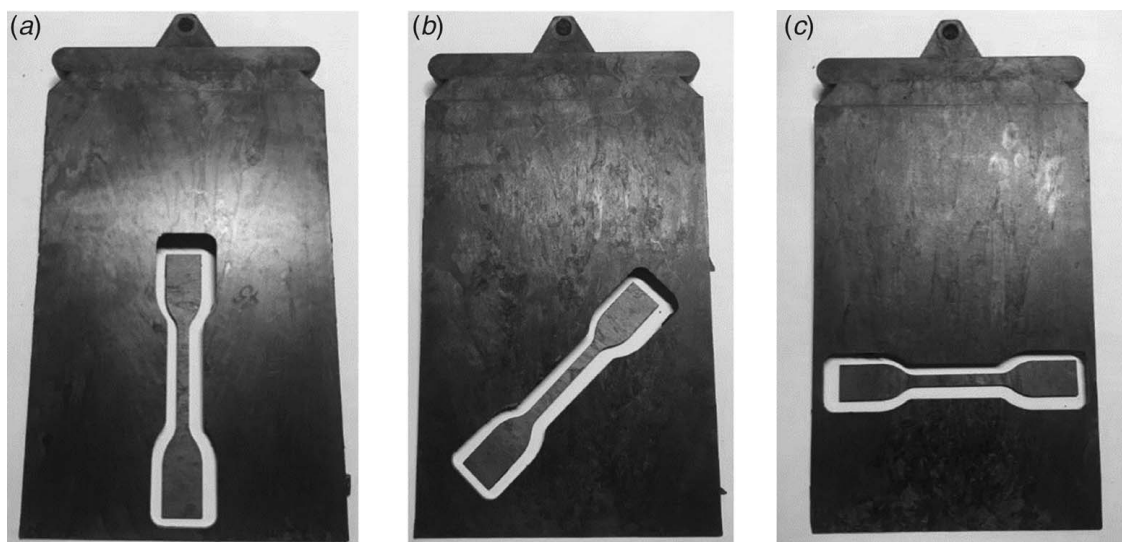


Fig. 1 Fabrication of (a) 0 deg specimen, (b) 45 deg specimen, and (c) 90 deg specimen

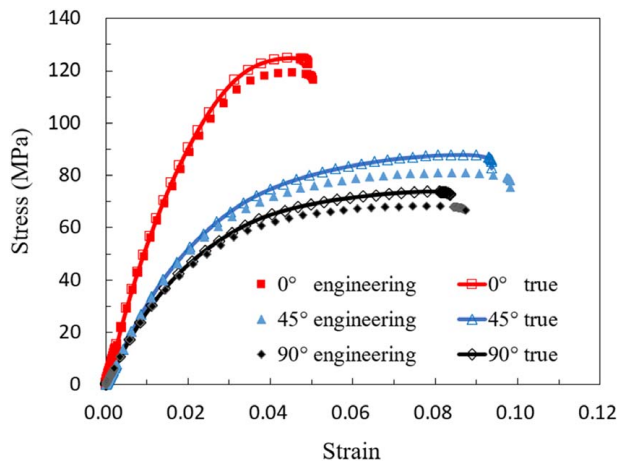


Fig. 2 Stress versus strain curves of the 0 deg, 45 deg, and 90 deg specimens

Table 1 Tensile properties of the 0 deg, 45 deg, and 90 deg specimen for PA66GF30 GFRP

	0 deg specimen	45 deg specimen	90 deg specimen
Elastic modulus, E (MPa)	5782.87	3225.44	3003.27
Yield stress, σ_y (MPa)	29.44	20.25	17.27
Tensile strength, σ' (MPa)	124.77	87.95	73.72
Failure strain, ϵ'	0.049	0.094	0.078

time of plate fabrication is about 10 s, which include the mold clamping time, injection time, hold time, and cooling time.

2.2 Specimens Preparation. Three different orientation ISO527 specimens are machined from injected PA66GF30 plates in 0 deg, 45 deg, and 90 deg directions concerning the molding flow direction, as shown in Fig. 1. The central position of each specimen is far enough from the injection gate and the plate side. To ensure the repeatability of the microstructure for the specimens that oriented in the same direction, only one specimen will be cut in each plate.

2.3 Tensile Test of the Specimens. To evaluate the mechanical properties of the PA66GF30 specimens, tensile properties were measured by using the Zwick/Roell machine according to ISO 527. In this test, all specimens were set by the standard conditions, which under the room temperature and 50% relative humidity. In contrast

with the dry molded state, the water absorption is relatively close to the actual working conditions.

Tensile specimens are equipped with a 25 mm extensometer to quantify the deformation in the center of the specimens. The test speed is 1.0 mm/min, five specimens are tested for each position, and the elastic modulus is determined by the slope between strain 0.0005 and 0.0025 of the stress–strain curve.

Figure 2 presents the corresponding stress–strain curves of the under-tension specimens. The indicated orientation angle refers to the orientation of the specimens with regard to the flow direction. Results show clearly the anisotropy via a strong effect of orientation on the Young's modulus and ultimate tensile stress.

Table 1 summarizes the elastic modulus E , the yield strength σ_y , the tensile strength σ' , and the failure strain ϵ' with the different direction specimens. For the specimens, the elastic modulus of 45 deg and 90 deg specimens are about 44% and 48% less than those of the 0 deg specimen, respectively. However, for the failure strain, the 45 deg specimen is measured with 9.4%, which is higher than both 90 deg specimen 7.8% and 0 deg specimen 4.9%.

3 Scanning Fiber and Microscopic Characteristics

3.1 Sample Preparation. The dimension of rectangular samples for μ CT scanning is 1.65 mm (thick), 4.0 mm (width), and 6.0 mm (length). The samples are machined from the injected PA66GF30 plates, as shown in Figs. 3(a) and 3(b), and the center position is exactly in the middle of the 0 deg direction specimen. Only one sample is cut per plate to ensure the repeatability of the microstructure.

3.2 Micro-Computed Tomography Scanning Measurement. To observe the orientation distribution of internal glass fibers, the NanoVoxel-4000 X-ray μ CT scanner is used to perform a microscopic scan on the 1.65 mm $\times$ 4.0 mm $\times$ 6.0 mm cuboid blocks. NanoVoxel-4000 X-ray three-dimensional (3D) μ CT is able to scan tomographic images with fine resolution that up to 1 μ m. The scanning sections of thickness $Z=0.075$ mm, $Z=0.475$ mm, $Z=0.785$ mm, $Z=0.865$ mm, $Z=1.185$ mm, and $Z=1.575$ mm are selected in Fig. 4.

The $Z=0.075$ mm and $Z=1.575$ mm, $Z=0.475$ mm and $Z=1.185$ mm, $Z=0.785$ mm and $Z=0.865$ mm are symmetric position, respectively, so in Figs. 5(a) and 5(f), 5(b) and 5(e), 5(c) and 5(d) fiber distributions are similar as well. In contrast to Figs. 5(a) and 5(c), the fiber direction varies about 90 deg. Comparing Figs. 5(a)–5(c), fiber on the different thicknesses with different fiber orientations is fully supported.

3.3 Orientation Tensor of Glass Fiber. Through X-ray μ CT scans, two-dimensional (2D) images could be obtained, and 3D images could be reconstructed by 2D image with specific software for observing in each axial direction. The microstructure of the μ CT sample is reconstructed by using VG studio. The orientation tensor

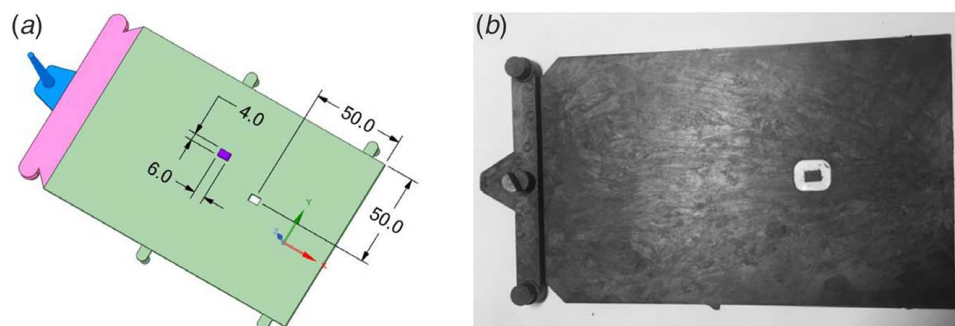


Fig. 3 Fabrication of (a) cutting blank and (b) μ CT sample (unit: mm)

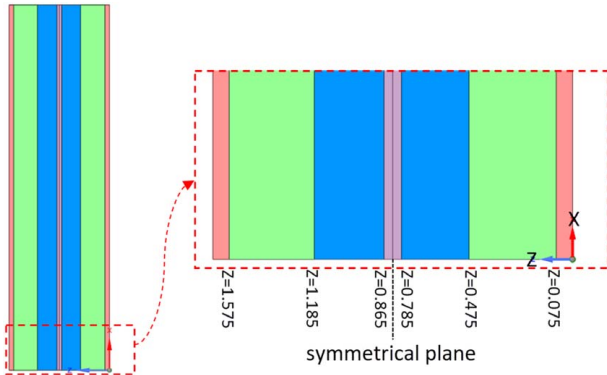


Fig. 4 Section position for the scanning object

of fiber in different thickness positions is combined and divided into the skin layer and core layer [42] along the thickness direction in Fig. 6.

4 Integrated Establishment of the Anisotropic Model of GFRP

4.1 Mean-Field Homogenization

4.1.1 Two-Step Mean-Field Homogenization. It is assumed that the composite is homogeneous on the macro scale, and the function of each component is only used to show the average properties of the composite. Material parameters in macro mechanical analysis, such as elastic modulus and tensile strength, can be obtained through the macro tensile test, as mentioned in Sec. 2.3. It can also be calculated by MFH analytical calculation based on the material properties of inclusion fiber and matrix. The purpose of MFH is computing approximately but accurately estimate of the volume averages of the stress and strain fields, and both are at the RVE level (macro stresses and strains) and in each phase.

To model short-fiber composites, the micromechanical Mori-Tanaka (M-T) model has been extended as a two-step incremental MFH formulation by Kammoun et al. [33,35] and Doghri and Tinel [38]. The two-step MFH is used to handle the mean-field homogenization of multiphase elastic-plastic materials reinforced with non-spherical and nonaligned inclusions. The two-step incremental MFH process is performed by decomposing RVE into an aggregate of pseudo-grains and homogenizing each pseudo-grain separately, then homogenizing the aggregate of homogenized pseudo-grains.

In the first step of the two-step procedure, each pseudo-grain occupies a diagonally symmetric effective stiffness predicted by M-T algorithm; the homogenization of each pseudo-grain is performed using a Hill-type incremental formulation. A fictitious reference material is defined for each phase, such that the rate (or incremental) stress-strain relations can be written as follows [35,43]:

$$\dot{\sigma}(x, t) = \hat{c}_r(t) : \dot{\epsilon}(x, t) \quad \forall x \in \omega_r, \quad r = 0, 1 \quad (1)$$

Where the reference tangent moduli $\hat{c}_r(t)$ are uniform per phase but vary with the time-like parameter (t). With a way to compute $\hat{c}_r(t)$, any homogenization model can be defined by its strain concentration tensors relating average strain rates in each phase to one another or to the overall (averaged over the pseudo-grain) strain rate. ω denotes the pseudo-grain's domain. The matrix (domain ω_0 inside the pseudo-grain) and inclusion (ω_1) phases are designated by indices (0) and (1), respectively.

In the second step, a homogenization scheme (e.g., Voigt), which guarantees the overall stiffness's symmetry will be chosen. At the end of the first step, we have computed the homogenized response of each pseudo-grain ω . During the second step, the final macro

response of the RVE (domain Ω) is obtained via homogenization of all pseudo-grains. RVE-average can be obtained as follows [35,43]:

$$\dot{\sigma}(x, t)_{\Omega} = c_{\omega_r}(t)_{i, \Psi_i} : \dot{\epsilon}(x, t)_{\Omega} \quad (2)$$

where $c_{\omega_r}(t)_{i, \Psi_i}$ represents the orientation distribution function (ODF) weighted average over all pseudo-grains in all phases i and all orientations. For isotropic linear elastic materials, each pseudo-grain is transversely isotropic, and a closed-form expression of c_{ω_r} exists which depends on its five invariants and the direction of anisotropy.

A two-step homogenization on the real microstructure does not give the same effective properties as a one-step homogenization of a two-phase composite with randomly oriented precipitates using M-T. The motivation behind preferring a two-step procedure to a one-step M-T is that for a linear elastic multiphase composite, diagonal symmetry of the overall stiffness tensor is guaranteed only when the dispersed inclusions have the same shape, aspect ratio, and alignment [44,45]. The two-step incremental MFH model has been implemented as a user-defined material law (UMAT) in the finite element code ABAQUS, and it is carried out in the framework of the DIGMAT software.

4.1.2 J_2 -Plasticity Model. The glass fiber is a homogeneous and isotropic linear elastic material, while the matrix PA66 is a homogeneous and isotropic elasto-plastic (EP) material.

The J_2 -plasticity model [46,47] of elasto-plastic constitution has been used for matrix PA66. This model is based on the von Mises equivalent stress σ_{eq} , defined as follows:

$$\sigma_{eq} = \sqrt{J_2(\sigma)} = \sqrt{\frac{3}{2} \mathbf{s} : \mathbf{s}} \quad (3)$$

where $J_2(\sigma)$ is the second invariant of the deviatoric stress tensor $\mathbf{s}$, and it is expressed as follows:

$$J_2(\sigma) = \frac{1}{2} [(\sigma_{11} - \sigma_{22})^2 + (\sigma_{22} - \sigma_{33})^2 + (\sigma_{33} - \sigma_{11})^2] + 3[\sigma_{12}^2 + \sigma_{23}^2 + \sigma_{31}^2] \quad (4)$$

For uniaxial loadings in formula (4), the von Mises equivalent stress is equal to the axial stress.

For the elasto-plastic constitution in this paper, the Cauchy stress and the elastic strain are related by

$$\sigma_{eq} = \mathbf{C} : \epsilon^e, \quad \sigma_{eq} \leq \sigma_Y \quad (5)$$

where $\mathbf{C}$ is Hooke's operator and ϵ^e is the elastic strain, and σ_Y is the initial yield stress.

When σ_{eq} exceeds the initial yield stress, the response becomes nonlinear and the plastic deformation appears. In this case, the Cauchy stress is given by

$$\sigma_{eq} = \sigma_Y + R(p), \quad \sigma_{eq} > \sigma_Y \quad (6)$$

where $R(p)$ is the hardening stress and p is the accumulated plastic strain. For uniaxial loadings in formula (6), the accumulated plastic strain is equal to the axial plastic strain.

The exponential and linear law is available for the hardening stress, which given by

$$R(p) = kp + R_{\infty}[1 - e^{-mp}] \quad (7)$$

where k is the linear hardening modulus, R_{∞} is the hardening modulus, and m is the hardening exponent.

The generalized reduced gradient (GRG) nonlinear algorithm is used to fitting test data to obtain that the σ_Y is 15.80 MPa, k is 8.06 MPa, R_{∞} is 56.51 MPa, and m is 91.67 for PA66 elasto-plastic constitution.

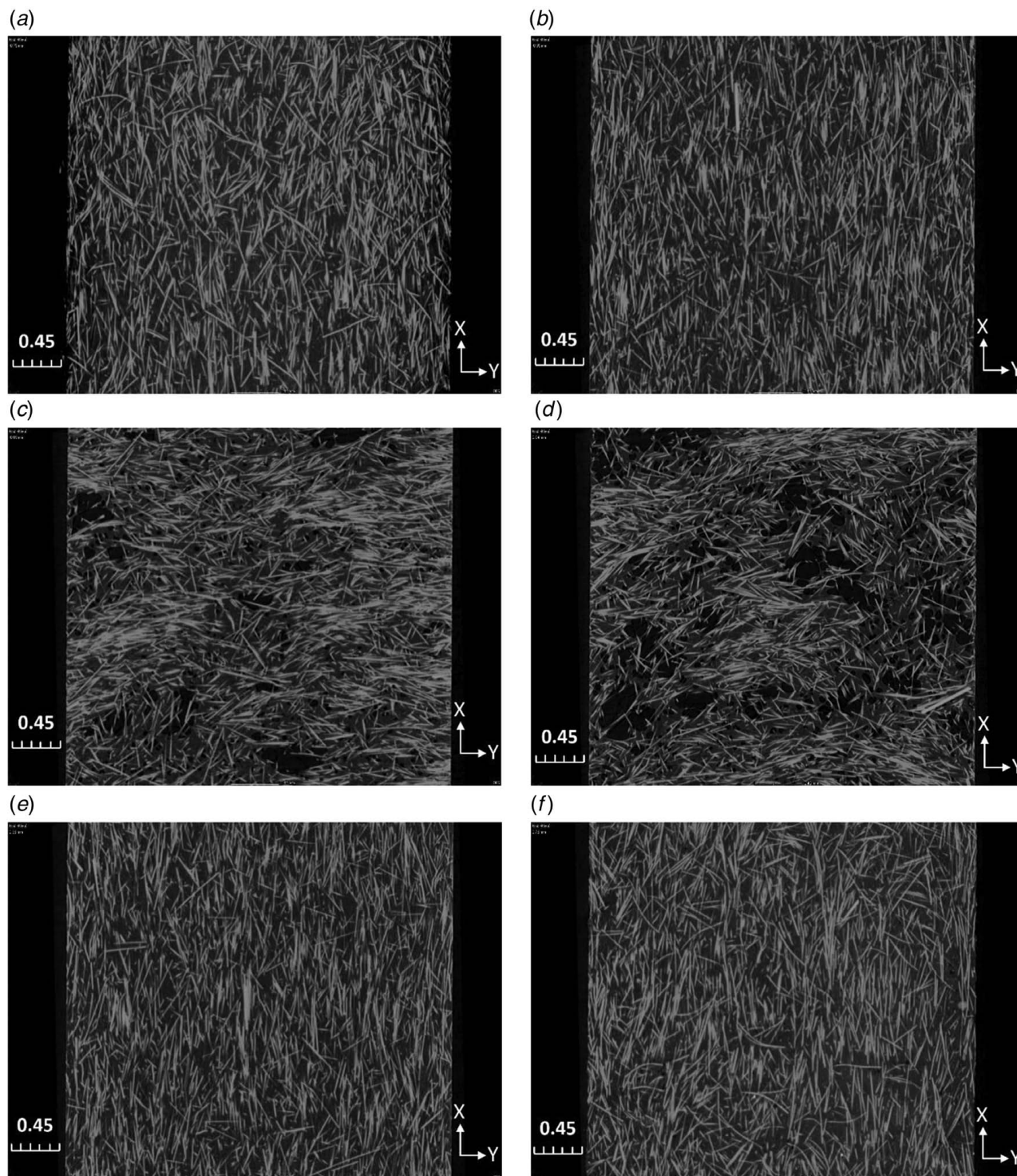


Fig. 5 Cross X–Y section image for different thickness positions at (a) $Z = 0.075$, (b) $Z = 0.475$, (c) $Z = 0.785$, (d) $Z = 0.865$, (e) $Z = 1.185$, and (f) $Z = 1.575$ (unit: mm)

4.1.3 Mean-Field Homogenization Results With PA66 and Fiber Parameters. Based on the fitting and μ CT measurement, the material and geometric parameters of the matrix PA66 and the inclusion fiber are summarized in Table 2.

In addition to the material and characteristic parameters listed in Table 2, the fiber orientation tensor in different thickness layers is also required for the analytic calculation of the two-step MFH algorithm. The curve data of Fig. 6 obtained by μ CT scan are interpolated to obtain the orientation tensor of the Z -center position of

20 layers fibers, listed in Table 3. In Table 3, since central position of Layer-1 and Layer-20 drop outside μ CT scanning range, the tensor of Layer-1 in Table 3 is the average value of the tensor Layer-2 and Layer-3, and the tensor of Layer-20 is the average value of the tensor Layer-18 and Layer-19.

Based on the parameters in Table 2 and the fiber orientation tensor in Table 3, two-step homogenizing calculation is performed to the 20-layer composites, and the strain–stress curves with the directions of 0 deg, 45 deg, and 90 deg are calculated, respectively.

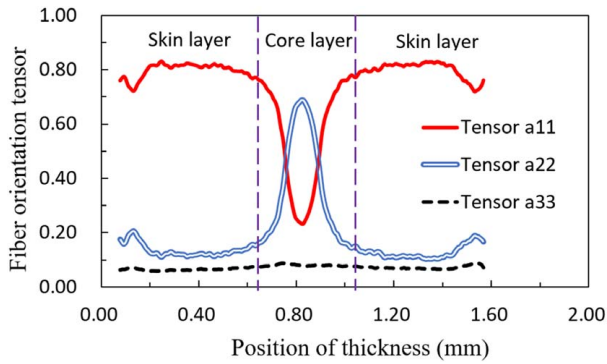


Fig. 6 Orientation tensor of fiber versus the position of thickness by μ CT measurement

Table 2 Parameters of PA66 and fiber for the GFRP

Parameters	PA66	Glass fiber
Material model	J_2 -plasticity	Elastic model
Young's modulus E (MPa)	1641.20	72,000.00
Poisson's ratio	0.35	0.22
Yield stress σ_Y (MPa)	15.80	—
Linear hardening modulus k (MPa)	8.06	—
Hardening modulus R_∞ (MPa)	56.51	—
Hardening exponent m	91.67	—
Mass factor (%)	70.0	30.0
Volume factor (%)	81.12	18.88
Fiber diameter D (mm)	—	0.016
Fiber length L (mm)	—	0.450
$AR = L/D$	—	28.125

The results are compared with the experimental curves and drawn in Fig. 7.

It could be found in Fig. 7 that the trend of the analytical value that calculated by the two-step MFH is consistent with the experimental value. However, the parameters are fixed value in the two-step MFH, and the parameters of each phase in the composite

Table 3 Fiber orientation tensors of 20-layer by interpolated μ CT scan data

Number	Thickness (mm)	Center position Z (mm)	Orientation tensors (μ CT scan)		
			a11	a22	a33
Layer-1	0.0825	0.041	0.772	0.164	0.064
Layer-2	0.0825	0.124	0.728	0.202	0.070
Layer-3	0.0825	0.206	0.816	0.125	0.059
Layer-4	0.0825	0.289	0.807	0.130	0.063
Layer-5	0.0825	0.371	0.820	0.117	0.063
Layer-6	0.0825	0.454	0.816	0.118	0.066
Layer-7	0.0825	0.536	0.800	0.131	0.069
Layer-8	0.0825	0.619	0.768	0.157	0.075
Layer-9	0.0825	0.701	0.705	0.215	0.080
Layer-10	0.0825	0.784	0.305	0.611	0.084
Layer-11	0.0825	0.866	0.322	0.599	0.079
Layer-12	0.0825	0.949	0.694	0.226	0.080
Layer-13	0.0825	1.031	0.779	0.146	0.075
Layer-14	0.0825	1.114	0.800	0.128	0.072
Layer-15	0.0825	1.196	0.816	0.115	0.070
Layer-16	0.0825	1.279	0.821	0.111	0.069
Layer-17	0.0825	1.361	0.827	0.106	0.067
Layer-18	0.0825	1.444	0.812	0.120	0.068
Layer-19	0.0825	1.526	0.726	0.186	0.088
Layer-20	0.0825	1.609	0.769	0.153	0.078

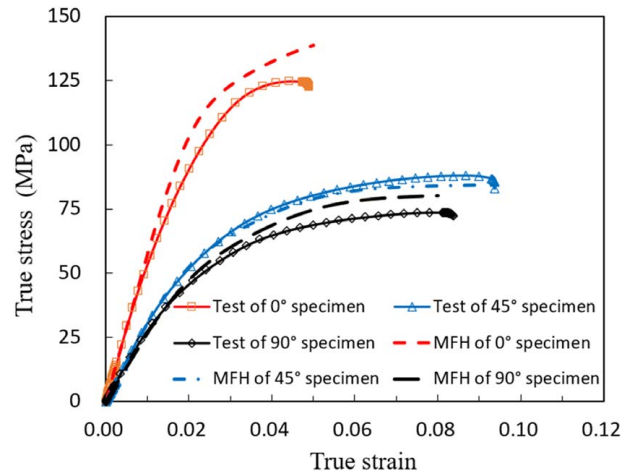


Fig. 7 Stress versus strain curves between two-step MFH and tensile test results

have a certain variation. In this case, there is still some deviation, and the deviation will be corrected by reverse iteration next.

4.2 Failure Indicators. Due to the anisotropy and inconsistency of local material properties, it is not reasonable to take a certain stress or strain value as the failure criterion. The failure indicator usually used as the criterion to judge the failure of materials. Failure indicators are real-valued functions while comparing given stress (strain) state combination to strengths (failure criteria). They are written in a normalized dimensionless form in such a way that it will be a safe state when an indicator value smaller than 1. On the contrary, failure is deemed to occur as soon as the chosen indicator is equal to or greater than 1.

The main failure criteria for fiber-reinforced composites are maximum stress or maximum strain, Hashin 3D, Tsai-Wu 3D, and Tsai-Hill 3D. Moreover, the Tsai-Hill 3D transversely isotropic strain-based failure criterion $F(\epsilon)$ is more applicable for fiber composite PA66GF30 [48–51], which is defined as for formula (8) by three maximum strain parameters without units:

$$F(\epsilon) = \frac{\epsilon_{11}^2}{X^2} - \frac{\epsilon_{11}(\epsilon_{22} + \epsilon_{33})}{X^2} + \frac{\epsilon_{22}^2 + \epsilon_{33}^2}{Y^2} + \left(\frac{1}{X^2} - \frac{2}{Y^2}\right)\epsilon_{22}\epsilon_{33} + \frac{(2\epsilon_{12})^2 + (2\epsilon_{13})^2}{S^2} + \left(\frac{1}{Y^2} - \frac{1}{4X^2}\right)(2\epsilon_{23})^2 \quad (8)$$

By assuming that the compressive strains equal to the tensile strains, this failure criterion has three continuous design variables (CDVs): the axial (X), in-plane (Y) maximum strains, and the transverse shear strain (S). The parameters X and Y must satisfy the relation: $Y < 2X$. This relation ensure that the material failure surface will form a closed envelope.

The failure indicators f of composite PA66GF30 could be expressed as formula (9):

$$f = \sqrt{F(\epsilon)} \quad (9)$$

According to the fracture characteristics of the tensile test curve, it can be considered that the fracture of composite material PA66GF30 is brittle fracture. Therefore, the problem of progressive stiffness degradation is not considered after the failure of the material in this paper. When $f \geq 1$, the stiffness degradation is directly defined as 0, and the corresponding failure element is deleted when the finite element analysis is used.

4.3 Optimization Parameters Using Reverse Engineering. The two-step MFH algorithm given in this paper is based on the matrix and inclusions of material performance and shape, while the

calculated strain–stress curves with the test curves have a certain deviation. To minimize calculation deviation, reverse engineering (RE) needs to be used on the two-step MFH algorithm to iteratively optimize the elastic modulus, yield strength, aspect ratio, and the parameters of failure indicators. Reverse engineering is an iterative optimization process aiming at finding the optimal values of design variables (i.e., material law parameters), which minimize the differences between model predictions and experimental measurements.

An optimization algorithm has been used to minimize the differences. Therefore, when doing a reverse engineering analysis, the first step is to formulate the optimization problem and to choose the optimization algorithm for solving it.

4.3.1 Optimization Problem Formulation. A mathematical formula for an optimization problem that expresses an estimate of error, which is hereafter called the error estimator e . When fitting the computed prediction curve to the reference test curve, the aim is to minimize the difference between those two curves. If the reference test curve is defined as a set of n (X_r, Y_r) points, the error estimator was defined as the relative error between the reference and computed curve

$$e = \frac{\sum_{i=1}^n (Y_c(i)/Y_r(i) - 1)^2}{n}, \quad i = 1, 2, 3, \dots, n \quad (10)$$

where $Y_c(i)$ and $Y_r(i)$ are the interpolated value of Y -ordinate of the computed curve and reference curve at X -ordinate equals to $X_r(i)$.

When several reference curves (let us say q curves) are provided, the optimization problem becomes a multi-objective optimization problem. This multi-objective optimization problem could be transformed to a single objective problem by using objective function weights, w_j . The global error estimator is then defined as the weighted sum of the error estimators

$$e_g = \sum_{j=1}^q w_j e_j, \quad j = 1, 2, 3, \dots, q \quad (11)$$

where e_g is the global error estimator, e_j is the error estimator of j curves, q is the number of the reference curve, and w_j is the objective function weight. The weight factor w_j of the objective function can be set according to the importance of the j curve.

Error estimator above can be used to optimize the elastic modulus, poison ratio, yield stress, hardening modulus, linear hardening modulus, hardening exponent, and aspect ratio that continuous design variables for the matrix PA66 and fiber. However, based on tensile break curves performed on the composite material with different loading angles, the parameters of the failure criterion are obtained or optimized by reverse engineering to determine the strengths of one failure criterion.

The optimization parameters of the failure criterion consist in the minimization of the relative square distance between the computed and experimental failure strains

$$\min [f] \text{ with } f = \frac{(\epsilon_f^{\text{comp}} - \epsilon_f^{\text{exp}})^2}{(\epsilon_f^{\text{exp}})^2} \quad (12)$$

where ϵ_f^{comp} and ϵ_f^{exp} are the computed and experimental failure strain, respectively. The strength objective function is defined to use the strain instead of stress to avoid the problem where the failure occurs on the plastic plateau of an elasto-plastic material.

Similarly, when several failure strains of curves are optimized, the weight factor p_j is introduced into formula (13) to transform it into a single objective optimization problem

$$f_g = \sum_{j=1}^r p_j f_j, \quad j = 1, 2, 3, \dots, r \quad (13)$$

where f_g is the global error estimators, r is the number of objective curves, and p_j is the objective function weight.

4.3.2 Optimization Algorithm. The used optimization method is the Dividing RECTangles (DIRECT) method [52] from the SCOLIB methods available in Dakota. The DIRECT optimization algorithm is a derivative-free global optimization method that balances local search in promising regions of the design space and global search in unexplored regions.

In practice, DIRECT has been proven as an effective heuristic for engineering design applications. It is able to quickly identify candidate solutions that can be further refined with fast local optimizers. First, the design space has to be defined, and this is done by specifying a search range for each CDV. Each CDV search space is bounded by an upper and lower bound. In DIGMAT, DIRECT simultaneously subdivides each subregion of the search space in all dimensions until one of the following criteria is satisfied: (1) the number of evaluations is larger than the parameter specified for the “maximum function evaluations.” Each computation of the global error estimator is considered as a function evaluation. By default, the maximum function evaluations are set to 200; (2) each subregion considered by DIRECT has a size, which corresponds to the longest diagonal of the subregion. The size of the smallest subregion falls below the threshold defined as the minimum box size. In practice, this latter specification is likely to be more effective at limiting DIRECT’s search. The default value for that threshold is 0.01 times the size of the design space.

4.3.3 Optimization Parameter Process. Based on the 20-layers fiber orientation, which is obtained from the μ CT scan in Table 3, the three-directional true stress–strain curves are taken as the reference curves in Fig. 2. The fiber and PA66 parameters in Table 2 are taken as the initial values for optimization, and the parameters are optimized by the DIRECT algorithm in reverse engineering with the two-step MFH model.

Reverse engineering will be performed in four stages as follows:

(1) Linear parameters of fiber

In addition to optimizing the elastic modulus E_{GF} of the fiber, it is also necessary to consider optimization of the AR of the fiber. The reason is that in the process of injection molding, the fiber will break at the top of the screw and inside the mold cavity due to shear load.

Setting fiber E_{GF} optimization in the range of $\pm 15\%$ and AR optimization in the range of $\pm 50\%$, the E_{GF} and AR parameters are optimized at strain ≤ 0.005 after 77 times iteration. The minimum global error optimized results have occurred in the 65th times iteration. The optimized E_{GF} and AR are 61,600 MPa and 14.24, respectively.

(2) Linear parameters of PA66

Based on the optimized linear parameters of fiber, the curves of strain ≤ 0.005 are optimized reversely under the elastic modulus E_{PA66} of PA66 in the range of $\pm 15\%$. After 21 times iteration, the 11th iteration results are the best, and the fitted value of E_{PA66} is 1434.5 MPa.

(3) Nonlinear parameters of PA66

Based on the optimized E_{GF} , E_{PA66} , and AR, the nonlinear parameters of matrix PA66, including σ_y , k , R_{∞} , and m are optimized iteratively at strain ≥ 0.005 . The range of increase and decrease of the parameters is set to $\pm 15\%$. After 137 times iterations, the fitting history is shown in Fig. 8(a), where the 128th iteration results are the best, and the optimized values of σ_y , k , R_{inf} , and m are 15.8 MPa, 8.06 MPa, 50.87 MPa, and 103.87, respectively. The stress–strain curves of the 128th optimization are drawn in Fig. 8(b).

(4) Constant parameters of failure criterion

In order to consider the failure characteristics of composite PA66GF30, based on the fitted three-directional stress–strain curves of Fig. 8(b) and the maximum tensile strain of the test curve, reverse engineering is carried out on the three constant parameters X , Y , and S of Tsai-Hill 3D transversely isotropic stain-based failure criterion.

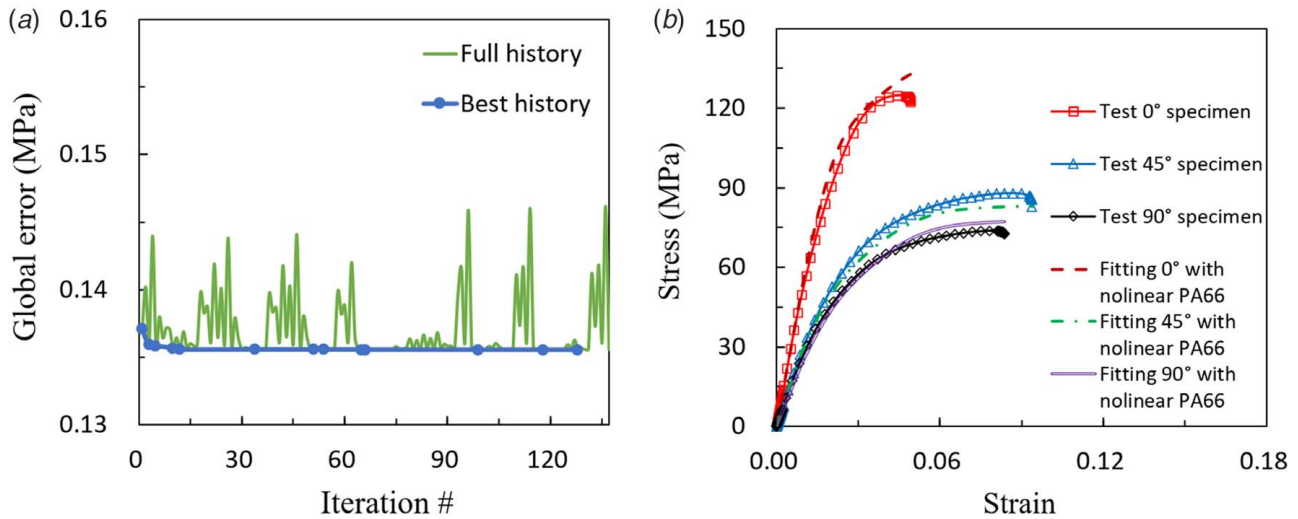


Fig. 8 (a) Global error and (b) fitting results for the nonlinear parameters of PA66

The initial values of X , Y , and S are set as 0.05, 0.09, and 0.3, respectively, and the range of these variations was limited to $\pm 25\%$. 108 iterations are optimized, and the optimal result is obtained at the 87th iteration. The X , Y , and S of the 87th optimization are 0.057, 0.095, and 0.367, respectively.

The above optimized ten parameters are as follows:

$E_{GF} = 61600$ MPa, $AR = 14.24$, $E_{PA66} = 1434.5$ MPa, $\sigma_y = 5.8$ MPa, $k = 8.06$ MPa, $R_\infty = 50.87$ MPa, $m = 103.87$, $X = 0.057$, $Y = 0.095$, and $S = 0.367$.

4.4 Integrated Establishment With Simulation and Mean-Field Homogenization

4.4.1 Injection Molding Simulation and Results. Micro-computed tomography scan can only obtain the fiber orientation of the local position on the plastic part, which is very time-consuming. However, using the injection molding simulation, the orientation tensor of the fiber at any position of the plastic part can be obtained quickly.

Based on the Fig. 1 plate and its mold, the finite element model (FEM) is established to simulate the injecting process and estimate the short-fiber distribution by using commercial software AUTODESK MOLDFLOW. To obtain the fiber orientation more accurately, the mesh in thickness is divided into 20 layers. In addition, the boundary conditions for the molding analysis are synchronized with the actual injection molding conditions with the single gate located at the top of the runner.

The reduced strain closure (RSC) model is introduced for predicting fiber orientations in MOLDFLOW. The RSC model is based on a scalar factor that reduces the growth rate of the eigenvalue of the orientation tensor while keeping the rotation rate of the eigenvalue of the orientation tensor constant. The RSC model includes two parameters of the fiber interaction coefficient C_i and reduced strain closure factor RSC_f . In this paper, the recommended value C_i and RSC_f from AUTODESK are adopted as 0.002 and 0.1.

The materials used in the injection molding simulation are consistent with the experiment, which is PA66GF30 HSBK161 from Kingfa Plastics (China) Ltd., and the parameters of non-Newtonian fluid behavior are tested by Autodesk laboratory (Australia), including the viscosity and specific volume of the material rheology. In addition, the specific heat and thermal conductivity are 2132 J/(kg $^{\circ}$ C) and 0.221 W/(m $^{\circ}$ C), respectively, at 290 $^{\circ}$ C.

The 20-layers fiber orientation tensor at the center of the plate is drawn from simulation results into Fig. 9 together with the μ CT test tensor. As can be seen from Fig. 9, the fiber orientation tensor predicted by the simulation is consistent with the μ CT scan test results

in the skin layers, and their distribution is symmetrical. It can be seen that the fiber orientation tensor a_{11} is highly variable in the thickness. On the other hand, the core layer showed a more complex fiber orientation distribution. At the same time, there is a certain deviation between simulation results and μ CT results in the core layer, which is caused by the limitation of MOLDFLOW software for the thin plate.

4.4.2 Tensile Specimen Simulation. The FEM of the specimen is created to predict the macroscopic mechanical behavior of the specimen, as shown in Fig. 10. The specimen geometry is segmented. The size of the segmented A zone of the specimen is same as that of the μ CT scan region, which means the X-direction distance between point p1 and point p2 is 6.0 mm.

The mesh of FEM consists of 296,000 elements with C3D8R type in 20 layers. The boundary conditions for the tensile simulation are synchronized with the actual tensile conditions. The top and bottom surfaces of the B zone are fixed in the model, the top and bottom surfaces of the C zone are coupled to the reference point (RP), and the displacement load is applied to the RP point in the X-direction.

According to the orientation tensor at the integral point of each element, the nonlinear behavior of each element is calculated by two-step MFH and applied for these elements of the FEM in Fig. 10. In Sec. 4.4.3, the orientation tensor of the integral points of each element is obtained from the result of injection simulation by the mapping method.

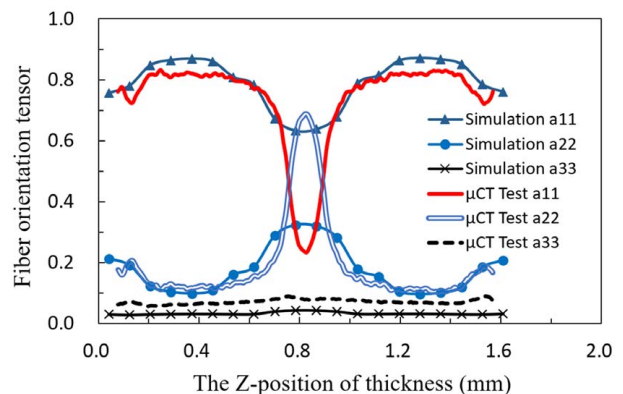


Fig. 9 The fiber orientation tensors from injection simulation and μ CT scanning measurement

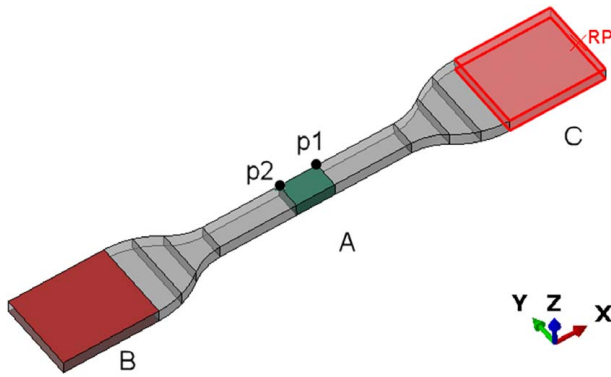


Fig. 10 The structural FEM of the specimen

4.4.3 Mapping Fiber Orientation. Since the finite element theory, the element type, and software used in injection simulation and structural simulation are different, the fiber orientation obtained in molding simulation is needed to transfer to the structure simulation model. The fiber orientation is represented by a 3×3 symmetric tensor, and marked a_{ij} is stored at the integration point of the element. Gaussian integral points in the element can represent the numerical distribution of the element domain accurately. Therefore, an integral point numerical mapping method is adopted in this paper.

The injection and structure FEMs in Fig. 10 use their own elements and types, respectively, and the integral points of the structural element are mapped and interpolated the required tensor value of fiber orientation. Before mapping, the position of the injection model is needed to adjust to the position of the corresponding structure model and both in the same coordinate system.

The structure model of the 0 deg specimen is aligned to the corresponding cutting position of the injection plate model, as shown in Fig. 11. The fiber orientation tensor of the specimen element of the structure model in Fig. 10 is obtained by mapping the integral point data. Through mapping data of the integral point tensor, the orientation tensor of the 0 deg specimen is obtained as shown in Fig. 12. It also can be seen that the fiber orientation tensor a_{11} is highly variable in the thickness.

Taking the Z-axis at the center of the specimen in Fig. 11 as the rotation axis, the aligned plate is rotated by counterclockwise 45 deg and 90 deg, respectively, to align with the corresponding cutting position of the specimen.

4.4.4 Predicted Nonlinear Behavior for the Specimen. Based on the fiber orientation of 0 deg, 45 deg, and 90 deg specimen mapped, as well as the two-step MFH algorithm with failure criteria

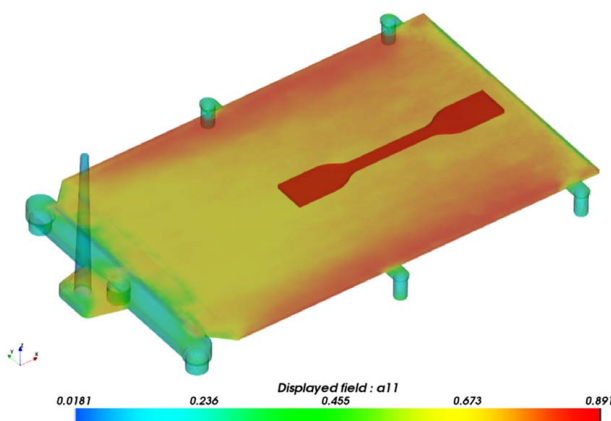


Fig. 11 The aligned 0 deg test specimen and injection plate model

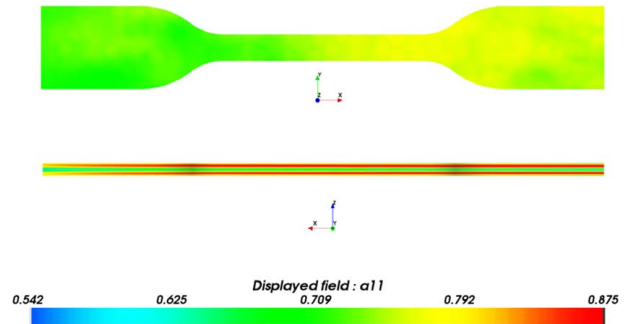


Fig. 12 The tensor a_{11} distribution of the 0 deg specimen elements

after optimized parameters, the mechanical behavior of the tensile specimen in Fig. 10 is solved by FEM.

The relative displacements between point p1, p2 and the cross-sectional forces have been extracted in Fig. 10. It combines together with the true strain and stress and the test results in Fig. 13. The comparison between simulation results and experiments in Fig. 13 fully illustrates that the adopted two-step MFH algorithm, the optimized material parameters, and the fiber orientation obtained by injection simulation can be used to predict the mechanical behavior of fiber composites accurately.

The maximum X-value of the 0 deg specimen curve in Fig. 13 is the moment of fracture state for the corresponding specimen, and the distribution of the failure indicator at this moment is shown in Fig. 14. Among them in Fig. 14, the fracture starts from the skin zone. The reason is that the fiber orientation in the skin zone is more consistent with the tension direction, and the deformation of matrix PA66 is not enough. In addition, due to the uneven fiber orientation on the specimen, the fracture location is not in the middle of the specimen, even that the fracture location is not in the range of extensometer gauge. It can be understood from the nonuniform distribution of the fiber tensor in Fig. 11 and can also be supplemented explanation from the fracture specimen in the tensile test.

The strain values of LE11 in the X-direction and strain LE22 in the Y-direction of 48,000 elements in the center A zone of the 0 deg specimen are listed in Fig. 15 and drop in the fourth quadrant. The (LE11, LE22) values are adjacent to the failure envelope, which indicates that the element is about to be failure. The element failure can be more intuitively understood by the distribution of failure indicators.

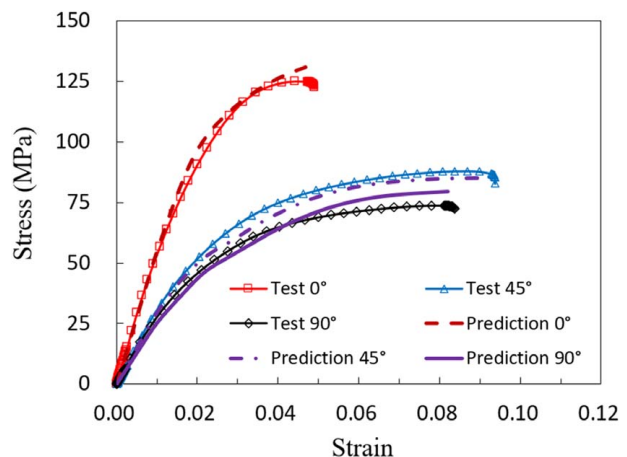


Fig. 13 Curves of strain versus stress between integrated prediction and test measurement

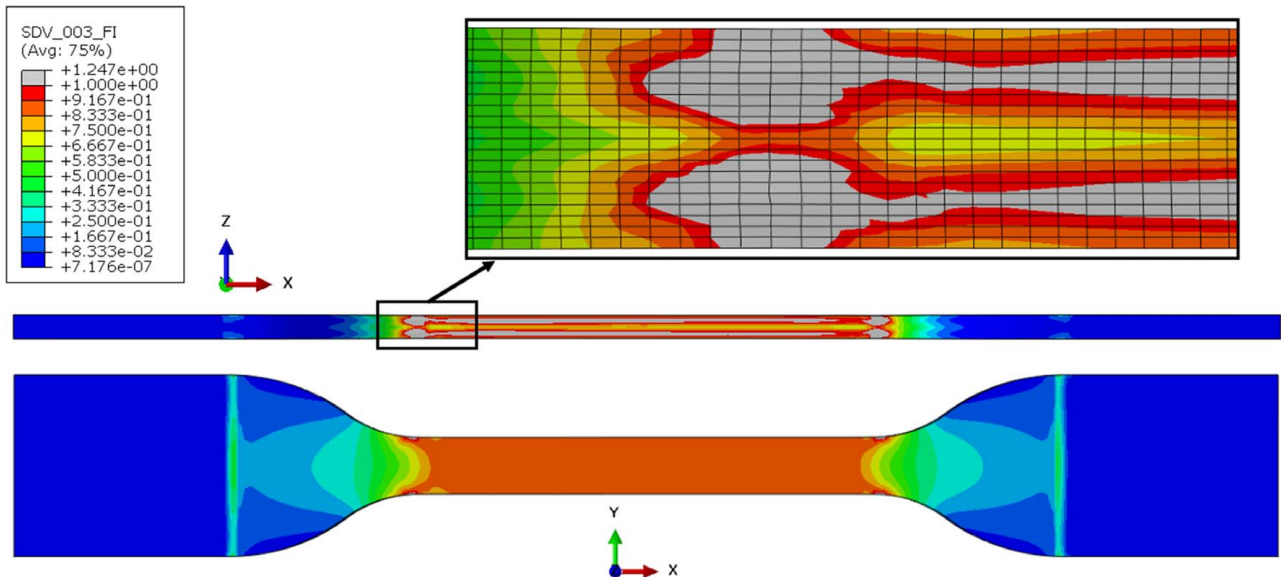


Fig. 14 The distribution of the failure indicator for the 0 deg specimen

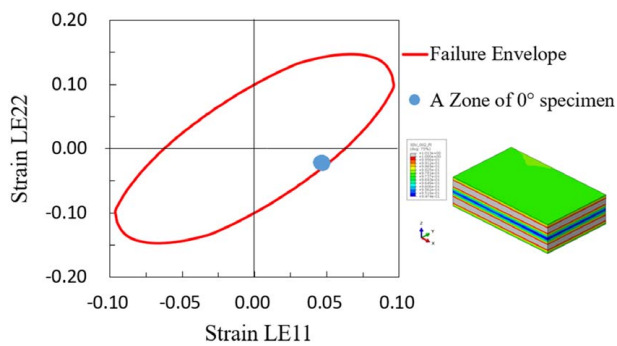


Fig. 15 Failure envelope and indicator distribution of A zone in the center of the 0 deg specimen

5 Experimental Verification for the Integrated Establishment

5.1 Ring Sample Preparation and Test. The ring sample is cut from the injection plate to verify the accuracy of the simulation prediction method in this paper. As shown in Fig. 16, two kinds of circular ring samples are processed in the X- and Y-directions, respectively. In order to save the injection plate and maintain the sampling consistency with the tensile specimen, the plate after-cutting specimen is used again.

The ring samples in Fig. 16 are tensile tested by the CTM8020 testing machine. The results of the tensile test will be described in Sec. 5.4.

5.2 Finite Element Model for the Ring Sample. For the ring samples in Fig. 16, the structure model is created, respectively, in Fig. 17. The structure model has 2,100,000 C3D8 elements with 20-layers mesh, and the mesh in the middle region of the ring is refined. The boundary conditions for the sample simulation are synchronized with the actual tensile conditions. The gripper region at one end is fixed, the gripper region at another end is constrained by the equation to the reference point, and the reference point will be pulled by 5.0 mm.

5.3 Mapping the Fiber Orientation Tensor. The structure model of the ring sample in Fig. 17 is aligned to the corresponding position of the injection model, as shown in Figs. 18(a) and 18(b). The integral point mapping method is used to map the data of the structure model from injection simulation.

5.4 Mechanical Behavior Prediction for the Ring Samples. Based on the fiber orientation of ring samples mapped in Fig. 18, as well as the two-step MFH algorithm with failure criteria after optimized parameters, the mechanical behavior of the tensile samples is solved by the FEM method. The solved force–displacement curves

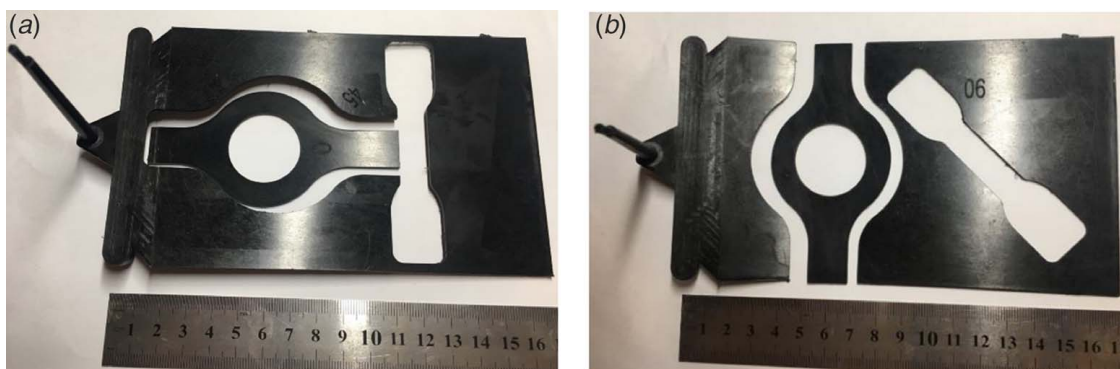


Fig. 16 Circular ring samples (a) in the X-direction and (b) in the Y-direction

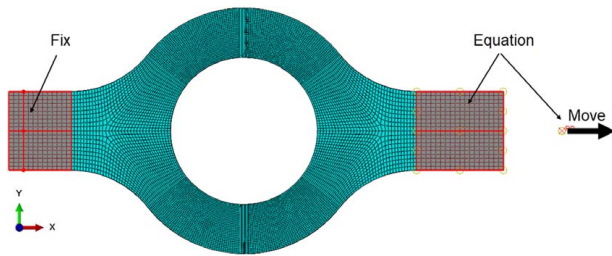


Fig. 17 Structure simulation model for circular ring

of tensile samples are plotted together with the test curves into Fig. 19.

As can be seen from Fig. 19, the maximum displacement obtained by simulation and test is consistent relatively. At the same time, the simulation prediction force–displacement curve for the Y-direction ring sample is consistent with the test curve; however, the simulation prediction force–displacement curve for the X-direction ring sample is slightly different from the test curve. This result shows that the predicted method has high accuracy in the field of the composite.

Figure 20 shows the predicted results that the distribution of the failure indicator at the initial fracture of the sample. The gray zone in Fig. 20 indicates that the value of the failure indicator is greater than 1, which indicating that these elements are close to failure, and they will be dismissed after further deformation to form a fracture gap.

As shown in Fig. 20(a), the fracture of the X-direction ring sample starts from the skin layer of Layer-2 and Layer-4, extending to the rest of the skin layer and core layer gradually. Figure 20(b) shows that the fracture of the Y-direction ring sample starts from the core layer and gradually, extending to the rest of the skin layer. It is determined by the fiber orientation and the force direction.

6 Discussion and Conclusion

The microstructure of fiber composite, material constitutive, and finite element method are combined in this paper. Employing tests with numerical simulation methods to build the material model of PA66GF30 and research the prediction method of anisotropic nonlinear mechanical behavior affected by the injection process. The main discussion and conclusions are listed below:

- The tensile test is implemented on the ISO527 specimens in 0 deg, 45 deg, and 90 deg directions to obtain the

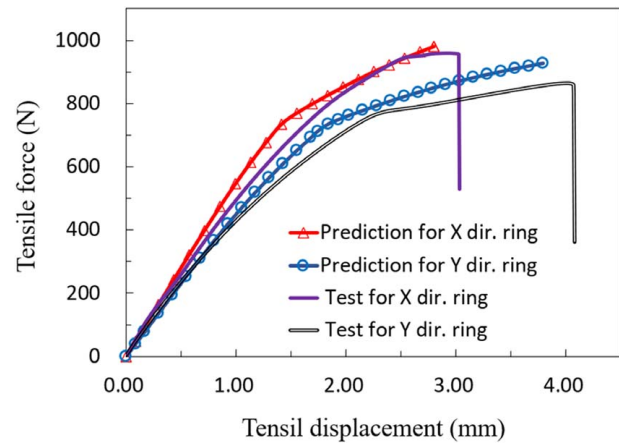


Fig. 19 Tensile curves for ring samples between tensile test and integrated prediction

corresponding stress–strain nonlinear curves. The results show obviously that the mechanical behavior of composite is anisotropic. Moreover, the elasticity modulus and tensile strength become larger with that angle between fiber direction and tensile direction gets smaller.

- Due to a large flow shear force on the surface near the mold cavity wall, the fibers oriented along the flow direction show a high consistency in the skin layer, while the flow shear force in the core layer is weak and the fiber orientation consistency is low [42]. The injected plate exhibits a classical skin-core microstructure, and the fiber orientations have highly skin-core layer characteristics and symmetric rules.
- From the μ CT scanning results, the core layer is about 0.40 mm-thick and its fiber orientation is mainly perpendicular to the injection direction; the fiber orientation of two symmetrical 0.625 mm-thick skin layers are mainly parallel to the injection direction. Also, it can be intuitively known from Figs. 9 and 12 that the orientation tensor of 20-layer fiber is highly inconsistent, showing obvious skin-core layer characteristics, and the core layer showed a more complex fiber orientation distribution.
- The fiber orientation tensor predicted by the simulation is consistent with the μ CT scan test results in the skin layers. However, it can be seen in Fig. 9 that there is a certain deviation between simulation results and μ CT results in the core layer, which is because the limitation of MOLDFLOW software

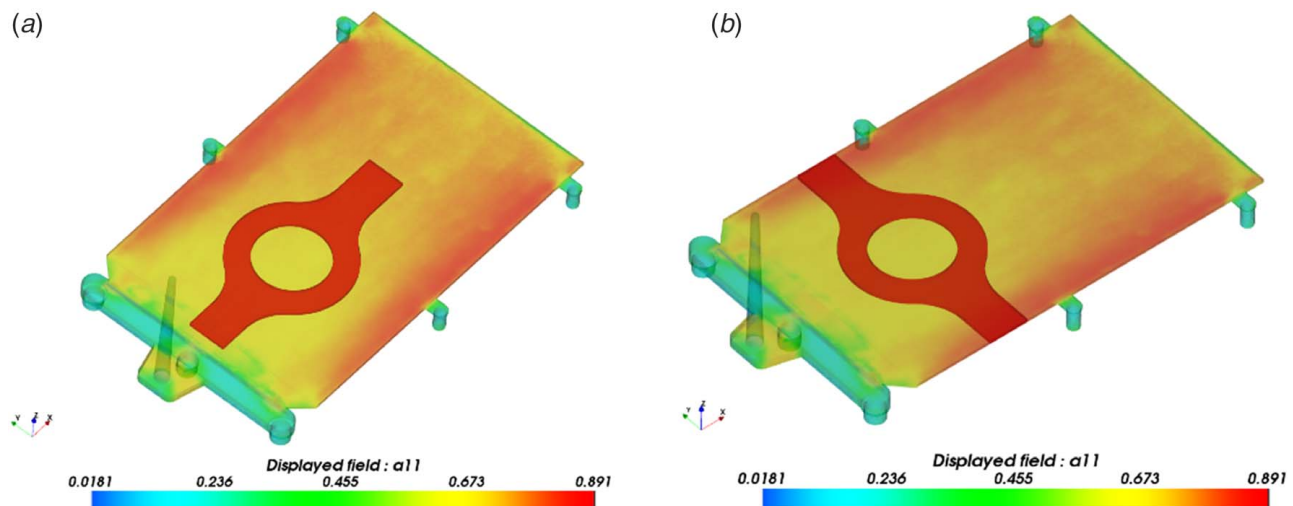


Fig. 18 Mapping fiber orientation tensor of circular ring samples in the (a) location of X-circular ring and (b) location of Y-circular ring

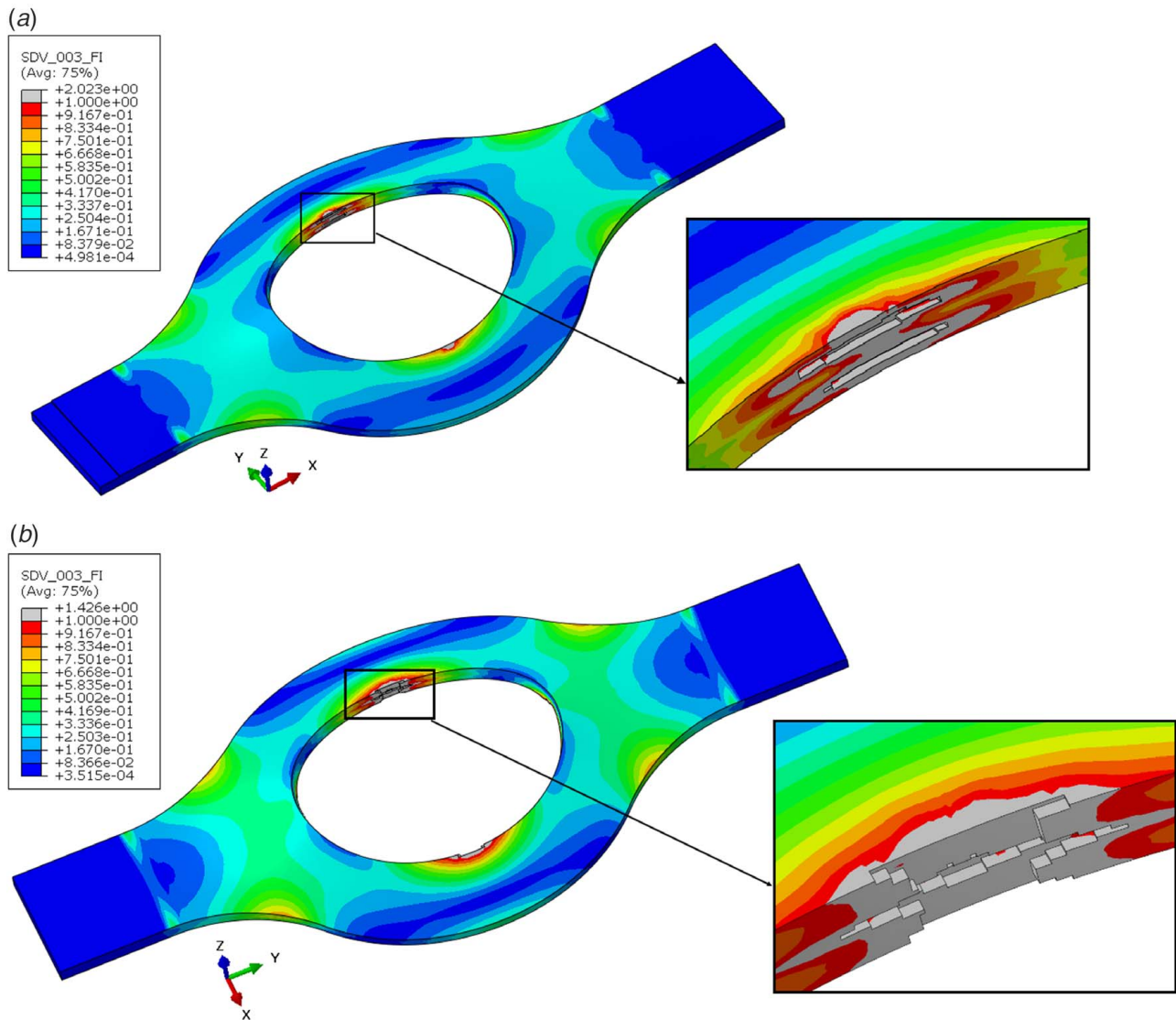


Fig. 20 The failure indicator distribution of the (a) X-direction ring sample and (b) Y-direction ring sample

for the thin plate. Luckily, the thickness of the core layer is only one-fifth of the injected plate. This deviation does not affect the prediction results too much.

- Based on the two-step MFH method, the optimized parameters of J_2 -plasticity and Tsai-Hill 3D transversely isotropic stain-based failure criterion are obtained through reverse engineering. Also, the fiber orientation tensor from injection simulation is transferred to the structure simulation model of the specimen through the integral mapping method. According to the known optimized parameters of J_2 -plasticity and failure criterion, the mechanical behavior of the specimen with 0 deg, 45 deg, and 90 deg directions are predicted, respectively. The prediction results show conformance with the real tests.
- According to the establishment of the prediction model and related parameters, the structure FEM of the X- and Y-direction ring samples are solved, respectively. The simulation results show the conformance with the test result. It means that the prediction method in this paper is practical and accurate.

To sum up, the test method and the integrated model establishment could provide the complete solution for the mechanical behavior prediction of fiber composites. Also, it could guide the precision and lightweight design of composite products.

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Conflict of Interest

There are no conflicts of interest.

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